Circles and Parabolas

ANSWER KEY

Circles

1 Write the standard-form equation of the circle with center $(3,\,-2)$ and radius 5.
$(x-3)^2+(y+2)^2=25$.

2 Consider the circle $x^2+y^2-6x+4y-12=0$.

a Write the equation in standard form by completing the square. $(x-3)^2+(y+2)^2=25$.

b State the center and radius. Center $(3,\,-2)$, radius 5.

3 Determine whether the point $(6,2)$ lies inside, on, or outside the circle $(x-3)^2+(y+2)^2=25$.
$(6-3)^2+(2+2)^2=9+16=25$, so the point lies on the circle.

Parabolas with vertex at the origin

4 For the parabola $y^2=8x$, find:

a the focus $(2,0)$

b the directrix $x=-2$

c the direction of opening opens right

5 A parabola has vertex at the origin, opens upward, and passes through the point $(4,2)$.

a Find its equation in the form $x^2=4py$. $16=4p(2)$ gives $p=2$, so $x^2=8y$.

b State the coordinates of the focus. $(0,2)$.

6 A satellite dish has a parabolic cross-section 2 m wide and 0.5 m deep. Taking the vertex at the origin opening upward, find the distance from the vertex to the focus (where the receiver is placed).
The edge point $(1,0.5)$ lies on $x^2=4py$: $1=4p(0.5)=2p$, so $p=0.5$ m. The receiver sits 0.5 m above the vertex.

Parabolas with vertex not at the origin

7 For the parabola $(y-2)^2=12(x+1)$, find:

a the vertex $(-1,2)$

b the value of p $4p=12$, so $p=3$

c the focus $(2,2)$

d the directrix $x=-4$

8 Write the equation of the parabola with vertex $(2,\,-3)$, opening downward, with $|p|=1$.
Opening downward: $(x-2)^2=-4(y+3)$.

9 Complete the square to write $y = x^2 - 6x + 5$ in vertex form, and state the vertex.

$y = (x-3)^2 - 4$. Vertex: $(3, -4)$.

Mixed practice: circles and parabolas together

10 A circle and a parabola are given by $x^2 + y^2 = 25$ and $y = x^2 - 5$. Find their points of intersection.

Substitute $x^2 = y + 5$ into the circle: $(y+5) + y^2 = 25$, so $y^2 + y - 20 = 0$, giving $(y+5)(y-4) = 0$: $y = -5$ or $y = 4$. For $y = -5$: $x^2 = 0$, point $(0, -5)$. For $y = 4$: $x^2 = 9$, points $(\pm 3, 4)$. Three intersection points: $(0, -5)$, $(3, 4)$, $(-3, 4)$.

11 A circular satellite orbit has the equation $x^2 + y^2 = (6400 + 500)^2$ (Earth radius plus altitude, in km). Find the radius of this orbit.

Radius $= 6400 + 500 = 6900$ km.